INCH-POUND

A-A-51181B
4 November 2005
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

GRINDING AND BUFFING MACHINES, UTILITY, BENCH AND PEDESTAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for electric motor driven, two-wheel, utility type, bench and pedestal mounted, grinding and buffing machines.

1.2 Intended use. The grinding machines covered by this CID are intended for grinding and buffing of tools and workparts which are held in the operator's hands. It does not cover cutting-off, cylindrical, internal, portable, saw gumming, snagging, or tool grinding machines.

2. CLASSIFICATION. The grinding machines shall be of the following styles and sizes:

2.1 Style. The style to be furnished shall be as specified (see 7.3(b)).

Style I - Bench mounted
Style II - Pedestal mounted

2.2 Size. The size to be furnished shall be as specified (see 7.3(c)).

Size 8 - 8-inch grinding wheel
Size 10 - 10-inch grinding wheel
Size 12 - 12-inch grinding wheel
Size 14 - 14-inch grinding wheel

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3415

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3. SALIENT CHARACTERISTICS

3.1 Requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustments, replacement, or repair. The grinding machine shall meet the requirements of American National Standards Institute (ANSI) B11.9, "Grinding Machines, Safety Requirements for the Construction, Care, and Use of", or shall be listed with Underwriter's Laboratories (UL) as meeting the requirements of UL 987, "Stationary and Fixed Electric Tools".

3.1.1 The grinding wheels and associated mechanisms shall meet the requirements of ANSI B74.2, "Specifications for Shapes and Sizes of Grinding Wheels, and for Shapes, Sizes, and Identification of Mounted Wheels", and ANSI B7.1, "Safety Requirements for the Use, Care, and Protection of Abrasive Wheels".

3.1.2 The machine shall have a grinding wheel mounted on each end of a spindle and shall meet the performance requirements shown in table I for the machine size indicated. Unless otherwise specified (see 7.3(d)), the machine shall include one coarse (36 grit) and one crimped wire (0.014 inch) wheel. Unless otherwise specified, the height (see 7.3(e)), and wheel-to-wheel centerline (see 7.3(f)) dimensions shall be in accordance with the manufacturer's standard practices.

TABLE I. Performance requirements.

Machine size (grinding wheel diameter)	Spindle speed (RPM)	Spindle motor rating (horsepower)
Size 8	3,450	3/4
Size 10	1,725	1
Size 12	1,800	2
Size 14	1,800	3

NOTE: The spindle speed and motor rating shall not be less than the values shown for the machine size indicated.

3.2 Electrical requirements. The machine shall operate from a dual 230/460 volt, 3 phase, 60 Hz circuit. Unless otherwise specified (see 7.3(g)), the machine shall be wired for 230 volt operation. The motor shall be energy efficient and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators".

3.3 Drill grinding attachment. When specified (see 7.3(h)), a drill grinding attachment shall be furnished with all necessary equipment for accurately grinding drill point angles from 90 degrees to 130 degrees on twist drills or flat drills having straight or taper shanks.

3.4 Wheel dresser. When specified (see 7.3(i)), a wheel dresser of the revolving cutter type shall be supplied.

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be met during the manufacturing, service, transportation, storage, and operation or use of the item.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

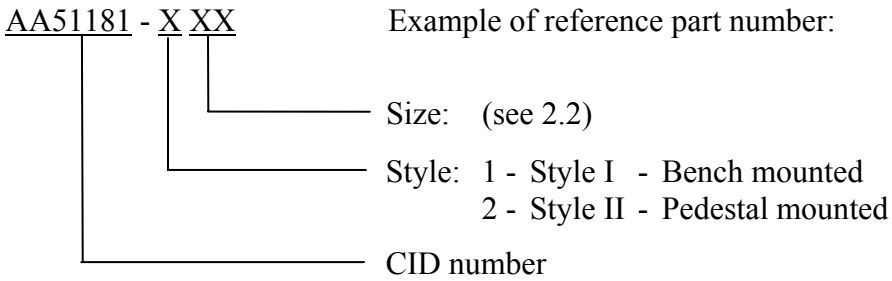
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(j)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



AA51181 - 110 indicates a bench mounted grinding machine with a 10-inch grinding wheel.

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7.2 Sources of documents.

7.2.1 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036. Electronic copies of ANSI standards may be obtained from http://web.ansi.org/.

7.2.2 UL standards. Copies of UL standards may be obtained from Underwriter Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096. Electronic copies of UL standards may be obtained from http://www.ul.com/.

7.2.3 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209. Electronic copies of NEMA standards may be obtained from http://www.nema.org/.

7.2.4 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.3 Ordering data. The acquisition order should specify the following:

a. CID document number, revision, and CID PIN.
b. Style of grinding machine (see 2.1).
c. Size of grinding machine (see 2.2).
d. Grinding and crimping wheels, if different (see 3.1.2).
e. Height of grinding machine (see 3.1.2).
f. Wheel-to-wheel centerline, if different (see 3.1.2).
g. Electrical requirements, if different (see 3.2).
h. Drill grinding attachment, if required (see 3.3).
i. Wheel dresser, if required (see 3.4).
j. Preservation, packing, and marking (see 6.1).

7.4 Subject term (key word) listing.

dresser
drill
wheel

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Air Force - 99

Review Activity:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

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